

Project Title

Smart Irrigation System Using ESP32 with Soil Moisture Sensor and Web Control

1. Introduction

Water is one of the most important resources for agriculture. Traditional irrigation systems often waste water due to manual control and lack of monitoring.

This project presents an **IoT-based Smart Irrigation System** using ESP32 that automatically controls water flow based on soil moisture level and also allows manual control through a web interface.

2. Objectives

- To automatically control irrigation based on soil moisture
 - To reduce water wastage
 - To provide both AUTO and MANUAL control modes
 - To enable remote monitoring and control using WiFi
 - To display real-time status on LCD
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3. Components Required

Component Name	Quantity
ESP32	1
Soil Moisture Sensor	1
Solenoid Valve / Relay Module	1
LCD Display (16x2 I2C)	1
Jumper Wires	As required
Power Supply	1

4. Working Principle

1. Soil Moisture Detection

The soil moisture sensor reads the moisture level of soil and gives an analog value. ESP32 converts this into percentage (0%–100%).

- Dry soil → Low moisture value
 - Wet soil → High moisture value
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2. Automatic Mode (AUTO)

When AUTO mode is enabled:

- Moisture < 30% → Motor ON (soil is dry)
- Moisture > 80% → Motor OFF (soil is wet)

This ensures efficient water usage.

3. Manual Mode (MANUAL)

In manual mode:

- User can control motor ON/OFF using web buttons
 - Full control is given to the user
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4. Web Server System

ESP32 creates a WiFi hotspot:

- Mobile or PC can connect
 - Real-time moisture monitoring
 - Mode switching (AUTO / MANUAL)
 - Motor ON/OFF control
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5. LCD Display System

LCD shows live system data:

- Current Mode (AUTO / MANUAL)
 - Soil Moisture Percentage
 - Motor Status (ON / OFF)
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5. Pin Configuration

Device	ESP32 Pin
Soil Moisture Sensor	GPIO 34
Solenoid / Relay	GPIO 26
LCD I2C SDA	GPIO 21
LCD I2C SCL	GPIO 22

6. Web Interface Features

- Real-time soil moisture update
 - AUTO/MANUAL mode switching
 - Motor ON/OFF control buttons
 - Auto-refresh every 1 second
 - Mobile-friendly UI design
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7. LCD Output Example

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Mode: AUTO  
M: 45%  
Mot: ON
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or

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Mode: MANUAL  
M: 70%  
Mot: OFF
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8. System Workflow

1. System starts and ESP32 creates WiFi hotspot
 2. Soil moisture sensor reads data continuously
 3. Data is converted into percentage
 4. System checks mode (AUTO or MANUAL)
 5. Motor is controlled automatically or manually
 6. Data is displayed on LCD and Web Dashboard
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9. Features

- ✓ Automatic irrigation control
 - ✓ Manual override system
 - ✓ Real-time web monitoring
 - ✓ LCD display output
 - ✓ Low-cost smart farming solution
 - ✓ Water saving system
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10. Limitations

- Sensor calibration required for accurate results
 - Affected by soil type and environment
 - Not suitable for large-scale industrial farming without expansion
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11. Future Improvements

- Add cloud IoT platform (Firebase / Blynk)
 - Mobile app control system
 - Solar-powered irrigation system
 - Multiple field zone control
 - Weather-based smart irrigation
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12. Conclusion

This ESP32-based Smart Irrigation System provides an efficient and intelligent solution for agricultural water management. It reduces water wastage, improves automation, and allows remote control through a web interface.